

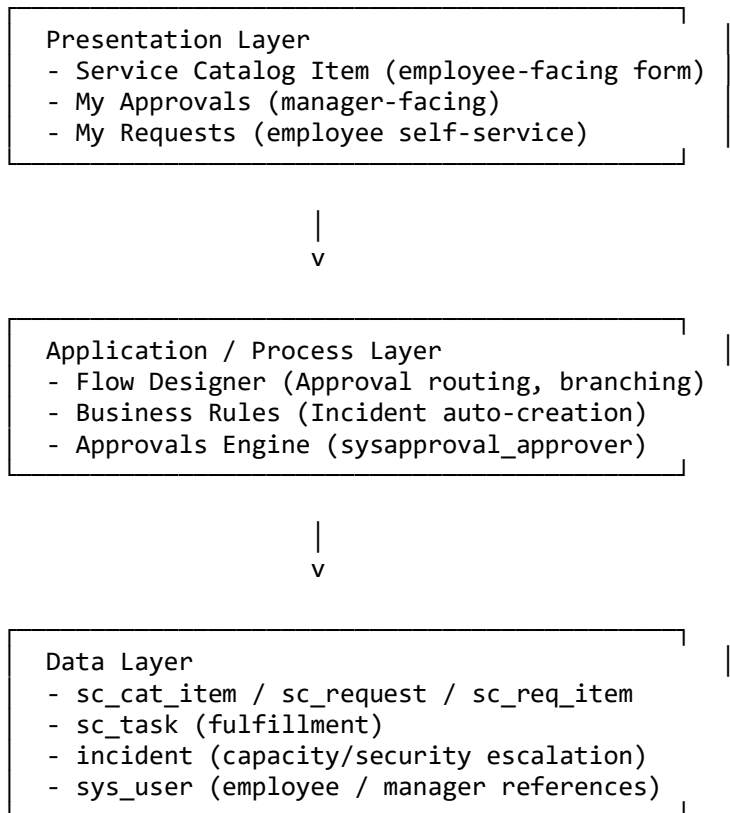
Solution Architecture

Project: Automated Network Request Management (ServiceNow)

1. Architecture Overview

This solution is built entirely on the ServiceNow platform using a scoped application, following a data-driven workflow architecture: a Service Catalog front-end feeds structured network access/change requests into standard Requested Item records, which drive an automated Approval and Fulfillment process via Flow Designer and Business Rules.

2. Layered Architecture



3. Component Interaction Diagram

Employee --(submits)--> Catalog Item

Catalog Item --(creates)--> Request --> Requested Item

Requested Item --(triggers)--> Approval Flow

Approval Flow --(assigns)--> Manager (My Approvals)

Manager --(decision)--> Approved / Rejected

Approved --> Catalog Task --> Network/IT Support Team

Rejected --> Email Notification --> Employee

On Hold (capacity/security conflict) --> Business Rule --> Incident
--> Network Support Group

4. Key Design Decisions

- Built as a scoped application to isolate all custom logic and enable clean packaging/version control.
- Used native Service Catalog + Approvals engine instead of custom tables, to leverage existing, well-tested ServiceNow infrastructure.
- Flow Designer (low-code) used for orchestration instead of classic Workflow Editor, for maintainability.
- Business Rules used only for simple, deterministic logic (e.g., Incident creation on a state change) — kept outside the main Flow to separate concerns.

5. Security Architecture

- Role-based access: Employee, Manager, and Network/IT Support roles define who can view/act on records.
- Approvals are restricted to the specific manager assigned via the Approval Field, preventing unauthorized approval actions.
- Cross-scope access explicitly granted only where required (e.g., email/notification creation from the scoped app).

6. Scalability & Extensibility

- Additional approval levels (e.g., network security team review) can be added as further branches in the Flow without redesigning the data model.
- A Network Capacity/Segment Inventory table can be introduced later to replace the simple On Hold/Incident escalation with real-time availability checks.
- The same architecture pattern can be reused for other request types (software, hardware, facilities access, etc.) by creating new Catalog Items pointing to the same Flow pattern.